DHRUV BELEKAR // FUTURE_MOBILITY

DRIVE

Reimagining driving for the visually impaired

An AI Co-Driver experience that brings freedom, agency and the joy of driving to everyone, everywhere.

AI CO-DRIVER

MULTISENSORY FEEDBACK

TACTILE INTERACTION

INCLUSIVE DESIGN

Driving is freedom.
Driving is control.
Driving is independence.

> But what if you've never experienced it?

Transportation is accessible.
Driving isn't.

ACCESSIBILITY LIMIT

Accessibility helps people reach destinations.

THE EXPERIENCE GAP

It rarely helps them experience the journey.

How might visually impaired individuals experience driving in an autonomous future?

Areas explored

L0 — No Automation

Complete human control. Visual and tactile environmental tracking are fully manual.

L1 — Driver Assist

Single automated safety systems. Tactile guidance and speed control act passively.

L2 — Partial

Combined steering and velocity control. Active monitoring is still human-dependent.

L3 — Conditional

System controls driving tasks. Human intervenes immediately when requested.

L4 — High Automation

Autonomous vehicle agency. Operates independently inside geofenced areas.

L5 — Full Automation

Zero driver inputs. The system operates autonomously under all roadway conditions.

> Key Insight: Autonomous vehicles solve mobility. They do not solve the experience of driving.

DRIVE: AI Co-Driver

A premium autonomous driving experience where AI becomes the eyes, sound becomes the road, and control becomes accessible to everyone.

01

Physical Interface

Tactile steering, mechanical shifters, and spring feedback pedals.

02

AI Companion

Real-time conversational partner translating speed, turns, and surroundings.

03

Environmental Soundscape

Synthesized engine rumbles and binaural audio cues orbiting the operator.

Eleven prototypes, three phases

The build moved from cardboard to digital fabrication to live electronics, testing one question at each step: does the driver feel the road?

PHASE 01 — CARDBOARD

Scale, proportion, ergonomics

01 Laser-cut frame

02 Cardboard prototype parts

03 Dashboard assembly

04 On-road user testing

PHASE 02 — DIGITAL FABRICATION

Structure and mechanism

05 3D printing: chassis

06 3D printing: haptic gears

07 Accelerator assembly

08 Responsive compression

PHASE 03 — FEEDBACK LOOP

Electronics, sound, haptics

09 Pedal electronics

10 Responsive testing

11 Haptic steering wheel

PHASE 01 — CARDBOARD · STEP 01 OF 11

Laser cutting frame

Rapid prototyping begins with laser-cutting cardboard profiles. This allowed us to quickly test scale, proportions, and mechanical ergonomics of the physical steering assembly.

Cardboard prototype parts

Laying out the early laser-cut cardboard wheels and linkage plates. This initial physical mockup established standard steering wheel diameter bounds and clearance limits.

PHASE 01 — CARDBOARD · STEP 03 OF 11

Dashboard assembly

Mounting the complete cardboard simulator assembly inside the test vehicle cabin. This allowed on-site ergonomics verification of steering height, reach parameters, and dashboard placement.

On-road user testing

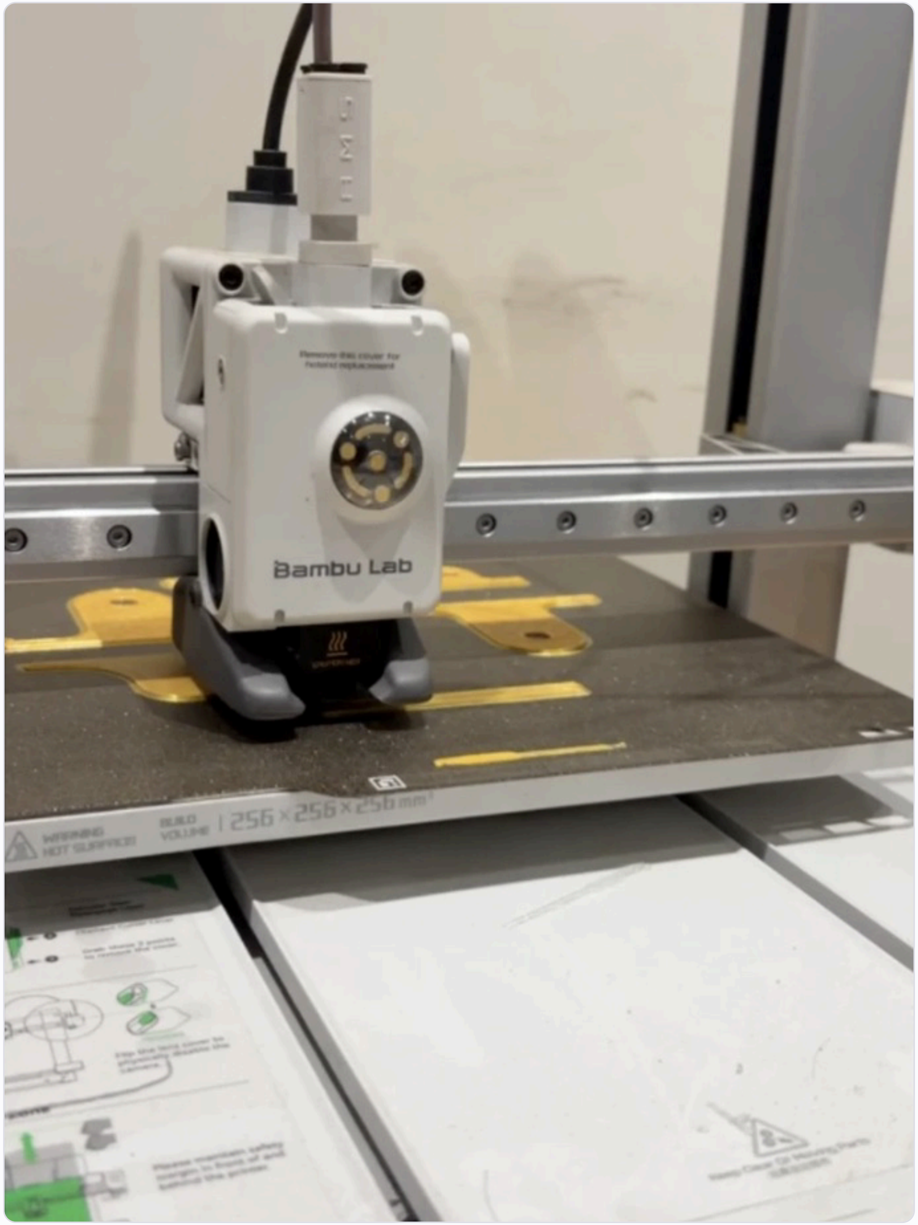
> "I want to feel like I am actually driving, not just being carried around as cargo."

Testing the prototype layout inside an actual vehicle cabin. Visually impaired users validated the critical need for active physical feedback rather than pure acoustic instructions.

PHASE 02 — DIGITAL FABRICATION · STEP 05 OF 11

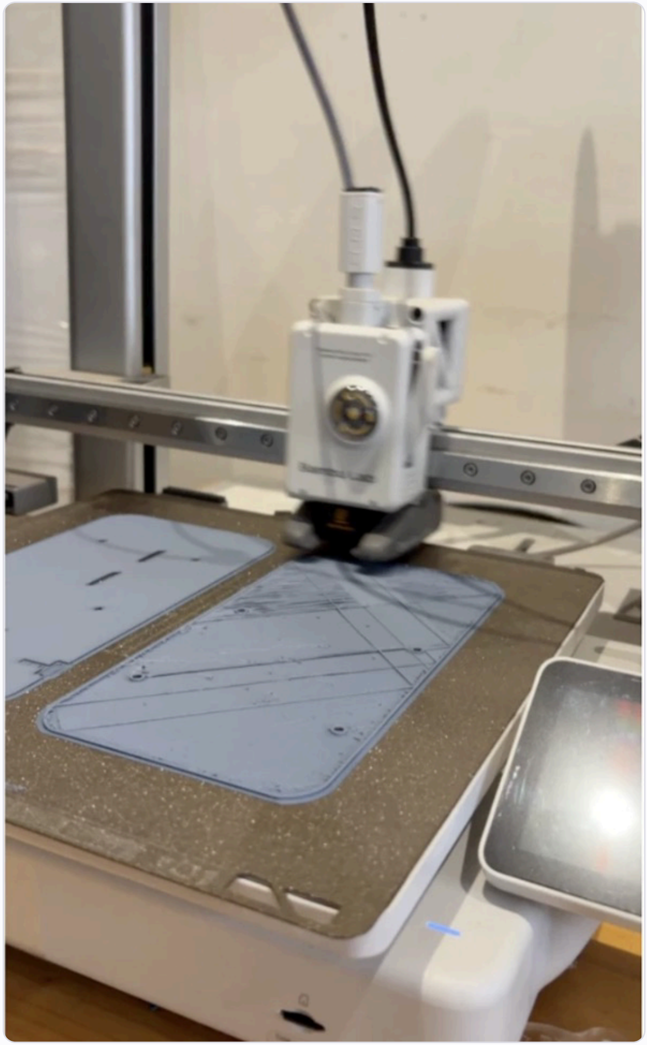
3D printing: chassis

Transitioning to digital fabrication. We utilized high-strength PLA to print the core structural chassis, creating robust housings for the internal electronics, sensors, and drive systems.



3D printing: haptic gears

Fabricating the internal gear train. Custom gears and brackets are printed using precision tolerances to transfer feedback directly from the steering motor to the driver's hands.



PHASE 02 — DIGITAL FABRICATION · STEP 07 OF 11

Accelerator assembly

The fabricated accelerator pedal chassis. Combining 3D-printed filaments with structural mounting elements, the pedal features integrated springs to replicate the physical travel resistance of standard automotive controls.

PHASE 02 — DIGITAL FABRICATION · STEP 08 OF 11

Responsive compression

Testing mechanical displacement. The spring-assisted compression allows for progressive travel, feeding physical position information back to the driver's foot.

Pedal electronics

Integrating pot sensors and an Arduino micro-controller. The electronics translate the physical angle of pedal travel into clean serial data streams to coordinate velocity changes with dynamic audio synthesis.

Responsive testing

Dynamic feedback in action. Pressing the pedal down changes the real-time simulation output, dynamically shifting the auditory rumbles and tactile pulses to correlate with speed levels.

INPUT

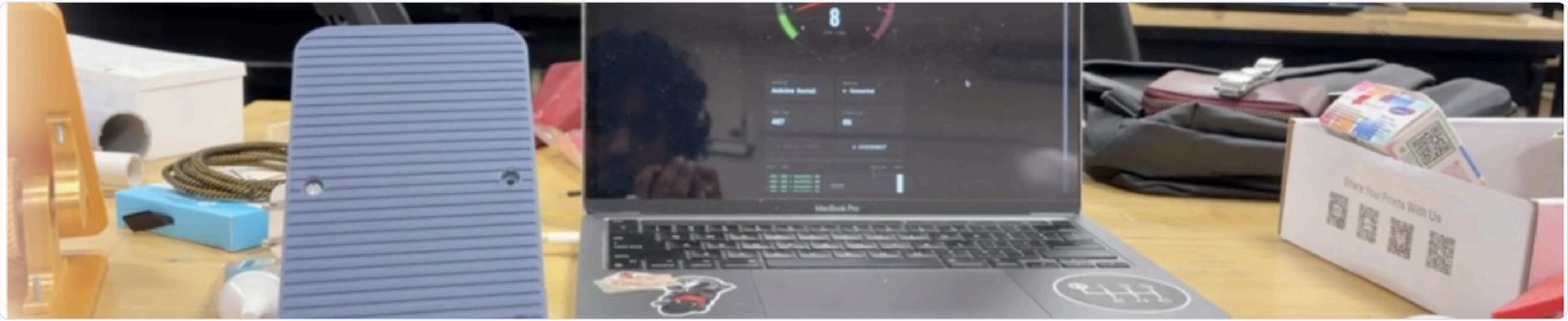
Pedal angle read by pot sensor

TRANSLATE

Serial stream to the simulation

OUTPUT

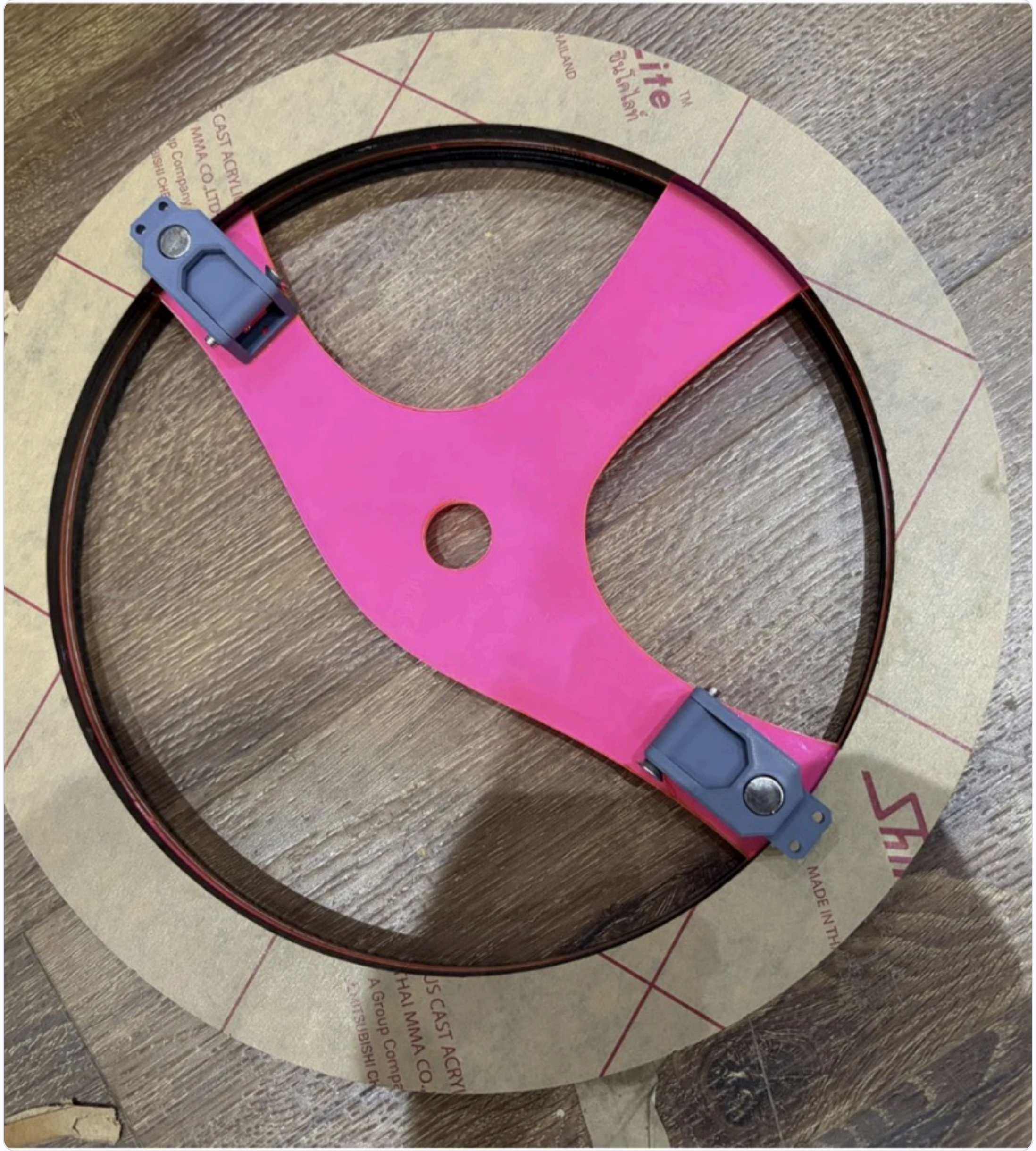
Rumble pitch and haptic pulse rate



PHASE 03 — FEEDBACK LOOP · STEP 11 OF 11

Haptic steering wheel

The final custom steering console assembly. Featuring integrated steering sensors, motors for mechanical torque assistance, and custom paddle shifters to provide the pilot a complete tactile grip of the drive environment.



The auditory narrative

During prototype testing, the co-driver drives the vehicle while the visually impaired operator sits in the passenger seat holding the prototype. A custom mobile sound box triggers binaural audio cues and spatialized alerts corresponding to physical milestones along the route.

01 // START

Welcome Driver

Onboard console pulses; Co-Driver AI greets operator.

02 // INITIATE

Engine Start

Tuning start dial releases steering lock; engine synthesizer rumble online.

03 // SPEED

Open Road

Wheel feedback shifts; spatial guide pitches align with curves.

04 // TUNNEL

Highway Tunnel

Audio reverb shifts; haptic signals tick as bridge pillars pass.

05 // CITY

City Limits

AI scans intersections; wheel detents indicate crosswalks.

"Accessibility is not only about reaching a destination. It's about experiencing the road along the way."

DRIVE

PROJECT

Feel of Driving for Visually Impaired

DESIGNED BY

Dhruv Belekar

DOMAIN

Future Mobility · Inclusive Design

EXPERIENCE THE ROAD. OWN THE JOURNEY.